At-Home HCT

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63.1 Introduction

The main indications for autoHCT are lymphoid malignancies (90%) with plasma cell disorders (MM and others) comprising 55% of all autoHCT (Passweg et al. 2021). Toxicity and mortality associated with autoHCT have been reduced significantly with the use of mobilized peripheral blood HSC, the extended use of cryotherapy associated with MEL, and the improvements in prophylactic antibiotic and antiemetic regimens. Besides this, outpatient parenteral antimicrobial treatment has been proven feasible and safe, thanks to modern CVC and infusion devices. All these advances have led to the development of outpatient autoHCT programs, and several studies have demonstrated their feasibility and safety (González et al. 2021).

There are various reasons for transferring the support of the neutropenic phase of autoHCT to the ambulatory setting, including patient preference, reduced exposure to hospital microorganisms, better use of hospital resources, and cost-saving issues (Martino et al. 2020). In this model, however, patients experience time-consuming daily travel to the outpatient clinic for blood tests and physician checkups. "Hospital at home" is an alternative, designed to reduce hospital outpatient admissions by providing hospital equivalent care to patients in the home setting (Fernández Avilés et al. 2006; González-Barrera et al. 2023).

63.2 Ambulatory AutoHCT Models (Martino et al. 2020)

Complete outpatient program	
Conditioning regimen, HPC infusion, and management of the aplastic phase	Outpatient clinic
Delayed admission	
Conditioning regimen, HPC infusion, and management of the aplastic phase	Inpatient Early discharged (+1) and readmission (+5)
Mixed inpatient-outpatient	
Conditioning regimen and management of the aplastic phase	Outpatient clinics
HPC infusion	Inpatient
Early discharge outpatient	
Conditioning regimen and HPC infusion	Inpatient
Management of the aplastic phase	Outpatient clinics
Early discharge at home	
Conditioning regimen and HPC infusion	Inpatient
Management of the aplastic phase	At home

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63.3 Suggested Inclusion Criteria for Ambulatory AutoHCT

Patient	Age ≤ 65 years, recommended in newly created programs ECOG ≤2 No organ failure Recent documented infection with a proven secondary prophylaxis Absence of refractoriness to platelet transfusion Signed written informed consent
Transplant Center	Outpatient clinics available 24 h per day or bed reserved in the transplant unit Dedicated phone line 24 h × 365 days to allow patients or their caregivers to contact an expert physician of the transplant team
Disease	CR or PR before the autoHCT No symptomatic advanced disease
Caregiver	Availability of a suitable caregiver 24 h per day, 7 days a week
Home	Clean house Travel time from home to the hospital less than 60 min at rush hours

63.4 General Recommendations for At-Home AutoHCT

Dose of HPC and supportive care (such as management of nausea and vomiting, hydration, analgesic therapy) should not differ from recommended conventional autoHCT guidelines. Antimicrobial prophylaxis for outpatient autoHCT should not differ from that required for conventional procedure, although some authors have intensified it with intravenous antibiotics significantly reducing the incidence and severity of neutropenic fever (Rodríguez-Lobato et al. 2020a, b). Recently, the usefulness of G-CSF after HPC infusion has been questioned; indeed, some transplant groups have stopped using it, observing no changes in relevant transplant outcomes and avoiding potential adverse effects including its potential association with engraftment syndrome (Grosso et al. 2022). Finally, some studies suggest that for MM patients, the addition of primary prophylaxis with corticosteroids after autoHCT may minimize the incidence of febrile neutropenia and engraftment syndrome, which would increase the safety of outpatient management of these patients without compromising outcomes (Mossad et al. 2005; Rodríguez-Lobato et al. 2020a, b).

63.5 Most Frequent Reasons for Readmission (Ordered by Frequency)

- Fever persisting after 2 days of broad-spectrum antibacterial therapy
- Severe oral mucositis or gastrointestinal toxicity (WHO grade III or IV)
- Severe sepsis with organic failure
- Request of the patient (psychological distress) and low compliance
- Loss of caregiver support

63.6 Treatment of Fever in At-Home Setting

The use of empiric antibacterial therapy must follow international accepted guidelines. If fever occurs, empiric broad-spectrum antibacterial therapy should be initiated within 1 h of the clinical evaluation and as soon as the fever workup was completed. IV antibiotics should be preferred and chosen in the light of clinical and laboratory findings. After at least 6-h monitoring, hemodynamically stable patients without relevant clinical problems may be followed at home. Table 63.1 shows the different empiric antibiotic therapy that could be used at home.

Table 63.1 Empirical antibiotic therapy

Prophylaxis with quinolones	Empirical antibiotic therapy
No	Levofloxacin (PO or IV) Moxifloxacin PO Ciprofloxacin PO associated or not with amoxicillin/clavulanate Ciprofloxacin PO associated or not with linezolid PO
Yes	– IV ceftriaxone or piperacillin/tazobactam[a] or meropenem[b] associated with teicoplanin IV[c] if intense oral mucositis – If there is a high suspicion of CVC infection, add teicoplanin and an anti-GNB such as amikacin IV, and evaluate the CVC withdrawal – If allergic to beta-lactam, quinolones PO/IV associated with teicoplanin IV and amikacin IV

[a] Stable at room temperature so it can be administered at home by electronic intermittent infusion pump
[b] Accurate refrigeration to achieve adequate stability for home administration
[c] The first option at home would be teicoplanin once daily instead of vancomycin IV (twice a day). Other alternatives rarely necessary in the context of autoHCT are daptomycin IV or linezolid PO

63.7 Incidence of Readmission in Outpatient and At-Home AutoHCT

The incidence of readmissions is closely related to the experience of the group of professionals in outpatient or at-home management of complications and by the support infrastructure available in the hospital.

In patients with MM, usually conditioned with MEL, the lowest readmission rates have been reported (less than 20%) due to the low organic toxicity (Martino et al. 2020). They are clearly the best option when considering starting an outpatient or at-home autoHCT program.

In patients with NHL or HL usually conditioned with a more toxic regimen (BEAM or BEAC), there is a higher readmission rate, between 20 and 30% (Scortechini et al. 2014; Jaime-Pérez et al. 2021) and 80% (Faucher et al. 2012).

In the *Hospital Clinic* at-home autoHCT experience, the actualized readmission rate in MM and lymphoma patients is only of 2% and 1%, respectively (Rodríguez-Lobato et al. 2020a, b), thanks to the significant reduction of febrile neutropenia rate with the intensification of antibiotic prophylaxis (piperacillin/tazobactam in NHL and LH patients) and the addition of primary prophylaxis of engraftment syndrome with corticosteroids after autoHCT in MM patients, as well as the successful control of fever at home. The overall readmission rate in this experience is significantly lower (6.5%) in a series of 537 patients.

63.8 Quality of Life

Patients who receive autoHCT experienced various symptoms on different levels and at different frequencies after transplant. Patients and their caregivers need guidance and a planned and individualized education in the process to reduce their anxiety (Caliskan and Can 2022). Thus, Summers et al. (2000) reported significantly higher scores for emotional well-being and global QOL in outpatients, while Martino et al. (2017) indicated that the outpatient model neither improves nor impairs global patient QOL on the first 30 days after autoHCT. Usually, a good clinical outcome following autoHCT was associated with better QOL and greater satisfaction with care. In summary, the data published are limited and contradictory, so the QOL remains an area that requires further research.

63.9 Cost Data

The study of "real" costs of these ambulatory/domiciliary autoHCT programs is still to be carried out. In the absence of well-designed studies aimed at evaluating the "real" savings achieved with outpatient/at-home autoHCT programs, some authors cite direct savings between 10 and 50% (González et al. 2021), especially influenced by the release of hospital beds and low readmission rates.

63.10 At-Home AlloHCT

Some centers have published their data regarding ambulatory alloHCT using RIC/NMA and MAC regimens and have demonstrated safety, feasibility, and cost-effectiveness. After more than two decades of the first ambulatory alloHCT program, the experience in this modality is limited. However, in low- and middle-income countries, where the development of a traditional HCT unit is often unrealistic, ambulatory alloHCT emerges as an affordable, safe, and realistic option.

Key Points

- At-home autoHCT is feasible and safe with a good selection of patients.
- MM is the best indication.
- There are no randomized studies that clearly indicate which model is better than another, which is the real impact of ambulatory autoHCT in the QOL of patients and their caregivers, as well as what is its real cost.

References

Caliskan K, Can G. Determining the symptoms and coping methods of patients at home after hematopoietic stem cell transplantation. Support Care Cancer. 2022;30:5881–90.

Faucher C, Le Corroller Soriano AG, Esterni B, et al. Randomized study of early hospital discharge following autologous blood SCT: medical outcomes and hospital costs. Bone Marrow Transplant. 2012;47:549–55.

Fernández Avilés F, Carreras E, Urbano-Ispizua A, et al. Case-control comparison of at home to total hospital care for autologous stem-cell transplantation for hematologic malignancies. J Clin Oncol. 2006;24:4855–61.

González MJ, Urizar E, Urtaran-Laresgoiti M, et al. Hospital and outpatient models for hematopoietic stem cell transplantation: a systematic review of comparative studies for health outcomes, experience of care and costs. PLoS One. 2021;16:e0254135.

González-Barrera S, Martín-Sánchez G, Parra-Jordán JJ, et al. Feasibility of a hospital-at-home program for autologous hematopoietic stem cell transplantation. Transplant Cell Ther. 2023;29:111.e1–7.

Grosso D, Leiby B, Wilde L, et al. A prospective, randomized trial examining the use of G-CSF versus no G-CSF in patients post-autologous transplantation. Transplant Cell Ther. 2022;28:831.e1–7.

Jaime-Pérez JC, Hernández-Coronado M, Picón-Galindo E, et al. Results of a completely outpatient autologous stem cell transplant program for lymphoma patients receiving reduced-intensity conditioning. Leuk Lymphoma. 2021;62:1619–28.

Martino M, Ciavarella S, De Summa S, et al. A comparative assessment of quality of life in patients with multiple myeloma undergoing autologous stem cell transplantation through an outpatient and inpatient model. Biol Blood Marrow Transplant. 2017;24:608–13.

Martino M, Paviglianiti A, Memoli M, et al. Multiple myeloma outpatient transplant program in the era of novel agents: state-of-the-art. Front Oncol. 2020;10:592487.

Mossad S, Kalaycio M, Sobecks R, et al. Steroids prevent engraftment syndrome after autologous hematopoietic stem cell transplantation without increasing the risk of infection. Bone Marrow Transplant. 2005;35:375–81.

Passweg JR, Baldomero H, Chabannon C, et al. Hematopoietic cell transplantation and cellular therapy survey of the EBMT: monitoring of activities and trends over 30 years. Bone Marrow Transplant. 2021;56:1651–64.

Rodríguez-Lobato LG, Martínez-Roca A, Castaño-Díez S, et al. The avoidance of G-CSF and the addition of prophylactic corticosteroids after autologous stem cell transplantation for multiple myeloma patients appeal for the at-home setting to reduce readmission for neutropenic fever. PLoS One. 2020a;15:e0241778.

Rodríguez-Lobato LG, Martínez-Roca A, Moreno DF, et al. Impact of intensifying primary antibiotic prophylaxis in at-home autologous stem cell transplantation program for lymphoma patients. Leuk Lymphoma. 2020b;61:1565–74.

Scortechini I, Montanari M, Mancini G, et al. Conditioning regimen with BCNU, etoposide, cytarabine and melphalan plus amifostine for outpatient autologous stem cell transplant: feasibility and outcome in 97 patients with lymphoma. Leuk Lymphoma. 2014;55:1657–60.

Summers N, Dawe U, Stewart DA. A comparison of inpatient and outpatient ASCT. Bone Marrow Transplant. 2000;26:389–95.

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